

PYTHON FOR DATA SCIENCE PRACTICAL 5

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ROLL NO : S049

CLASS : SYIT

- ✓ 5A. Write a python code to demonstrate importing pandas libraries and create data frame object.

```
import pandas as pd

data = {
    'Name': ['Komal', 'Laxmi', 'Anjali'],
    'Marks': [85, 90, 78],
    'City': ['Mumbai', 'Pune', 'Nashik']
}

df = pd.DataFrame(data)

print(df)
print("Komal S049")
```

	Name	Marks	City
0	Komal	85	Mumbai
1	Laxmi	90	Pune
2	Anjali	78	Nashik

Komal S049

- ✓ 5B. Write a python code to show statistical information on given data set.

MEAN :

```
import statistics
print(statistics.mean([1, 3, 5, 7, 9, 11, 13]))
print(statistics.mean([1, 3, 5, 7, 9, 11]))
print(statistics.mean([-11, 5.5, -3.4, 7.1, -9, 22]))
print("Komal S049")
```

7
6
1.8666666666666667
Komal S049

MEDIAN :

```
import statistics
print(statistics.median([1, 3, 5, 7, 9, 11, 13]))
print(statistics.median([1, 3, 5, 7, 9, 11]))
print(statistics.median([-11, 5.5, -3.4, 7.1, -9, 22]))
print("Komal S049")
```

```
7
6.0
1.05
Komal S049
```

MODE :

```
import statistics

# Calculate the mode
print(statistics.mode([1, 3, 3, 3, 5, 7, 7, 9]))
print(statistics.mode([1, 1, -3, 3, 7, -9]))
print(statistics.mode(['red', 'green', 'blue', 'red']))
print("Komal S049")
```

```
3
1
red
Komal S049
```

✓ 5C. Write a python code to create pandas series from dictionaries.

```
import pandas as pd

data = {
    'Komal': 85,
    'Laxmi': 90,
    'Anjali': 78
}

series = pd.Series(data)

print(series)
print("Komal S049")
```

```
Komal      85
Laxmi      90
Anjali     78
dtype: int64
Komal S049
```

✓ 5D. Write a python code to demonstrate filter pandas series with Boolean arrays.

```
import pandas as pd

data = {
    "Name": ["Laxmi", "Shalu", "Komal", "Anjali", "Amatullah"],
    "Hobby": ["Reading", "Dancing", "Singing", "Painting", "Cooking"],
    "Qualified": [True, False, True, True, False]
}
```

```
df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

print("\nQualified Students:")
print(df[df["Qualified"] == True])
print("Komal S049")
```

Original DataFrame:

	Name	Hobby	Qualified
0	Laxmi	Reading	True
1	Shalu	Dancing	False
2	Komal	Singing	True
3	Anjali	Painting	True
4	Amatullah	Cooking	False

Qualified Students:

	Name	Hobby	Qualified
0	Laxmi	Reading	True
2	Komal	Singing	True
3	Anjali	Painting	True

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